Evaluate Expressions

Lesson 6-2

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

In $3x + 5$, x can be any number — if $x = 2$, the expression equals 11; if $x = 10$, it equals 35

Variable

A letter that stands for an unknown number.

$2n + 7$ is an expression: the $2n$ means '2 times some number' and the $+ 7$ means 'then add 7'

Expression

A math phrase with numbers and letters, but no equal sign.

Evaluate $2n + 7$ when $n = 4$: replace n with $4 \rightarrow 2(4) + 7 = 8 + 7 = 15$

Evaluate

To find the value by putting numbers in for the letters.

Substitute 3 for x in $5x$: replace x with $3 \rightarrow 5(3) = 15$

Substitute

To put a number in place of a letter.

In $7m$, the coefficient 7 tells you to multiply m by 7 — if $m = 3$, then $7m = 21$

Coefficient

The number in front of a letter, like the 3 in $3x$.

$4x$ and $2x$ are like terms (both x^1); $4x$ and $4y$ are NOT (different variables)

Like terms

Terms with the same letter, like $2x$ and $5x$.

Key Ideas & Notes

- You're a music producer setting up a concert.
- The total cost of the event is given by the expression $150 + 12s$, where s is the number of speakers rented.
- The venue charges $150 as a base fee, plus $12 for each speaker.
- How much will it cost for different numbers of speakers?
- Evaluate the expression $150 + 12s$ for each number of speakers.

Think About It

- What does the 150 represent in the expression?
- What does the $12s$ represent?
- What happens to the total cost as s gets bigger?

My Notes

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The concert cost is $150 + 12s$, where s is the number of speakers. How do you find the cost for 5 speakers, and why does this one expression work for any number of speakers?

Level 1 support

Start here: Substitute the number of speakers for s, then evaluate.

I substitute $s = 5$ into $150 + 12s$ to get ____.

Sustituyo $s = 5$ en $150 + 12s$ para obtener ____.

One expression works for any show because s ____.

Una expresión sirve para cualquier concierto porque s ____.

Word bank: Variable | Expression | Evaluate | Substitute

Listen for: *Students substitute $s = 5$ ($150 + 12 \times 5 = 210$), explain 150 is the fixed venue fee and 12 is the cost per speaker, and note s can be any number.*

Level 2

Predict whether doubling the speakers from 5 to 10 doubles the total cost. Justify using the $150 base fee.

- Doubling speakers does ____ double the cost because the 150 ____.
- For 10 speakers the cost is ____, which compared to 210 shows ____.

EXPLORE

When you evaluate an expression like $4x^2 + 3$, what order do you follow, and why must you square before you multiply?

Level 1 support

Start here: Follow order of operations: substitute, then exponents, then multiply, then add.

First I substitute, then I do ___ because of the order of operations.

Primero sustituyo, luego hago ___ por el orden de las operaciones.

I must square before multiplying because ___.

Debo elevar al cuadrado antes de multiplicar porque ___.

Word bank: **Evaluate** **Substitute** **Expression** **Variable**

Listen for: *A strong answer substitutes the value first, then applies the exponent before multiplying (e.g., $4 \times 2^2 = 4 \times 4 = 16$, not $(4 \times 2)^2$), citing order of operations.*

Level 2

A classmate evaluates $4x^2 + 3$ at $x = 2$ as $(4 \times 2)^2 + 3 = 67$. Critique the error and give the correct value.

- The error is ___ because the exponent applies only to ___.
- The correct value is ___ because I do 2^2 = ___ first, then ___.

CONNECT

The rideshare fare is 2.50 + 1.75m for m miles. How much is an 8-mile ride, and what does each part of the expression represent?

Level 1 support

Start here: Substitute m = 8 and evaluate to get the total fare.

I substitute m = 8: 2.50 + 1.75 × ____ = ____.

Sustituyo m = 8: 2.50 + 1.75 × ____ = ____.

The 2.50 is the ____ and the 1.75 is the ____.

El 2.50 es la ____ y el 1.75 es la ____.

Word bank: Substitute | Variable | Evaluate | Expression

Listen for: *Students evaluate 2.50 + 1.75 × 8 = 16.50 and identify 2.50 as the base fee and 1.75 as the per-mile rate.*

Level 2

Two riders pay the same fare, but one traveled fewer miles. How is that possible with 2.50 + 1.75m? Justify what would have to differ.

- That is possible only if ____, because the per-mile rate 1.75 ____.
- The base fee 2.50 is the same for both, so any difference comes from ____.

Guided Examples

Example 1

Evaluate $3x + 7$ when $x = 4$.

Solution: $3(4) + 7 = 12 + 7 = 19$.

Answer: A. 19

Example 2

Evaluate $20 - 2n$ when $n = 6$.

Solution: $20 - 2(6) = 20 - 12 = 8$.

Answer: A. 8

Example 3

Which expression means '5 less than twice a number n'?

Solution: Twice n is $2n$. Five less than that is $2n - 5$.

Answer: A. $2n - 5$

Write About the Math
The Writing Revolution

I can explain my steps using the words variable, expression, evaluate, and substitute.

1. Kernel Sentence
subject + verb

Model: Expression is a math phrase with numbers and letters, but no equal sign.
Expresión es una frase matemática con números y letras, pero sin signo igual.

Write a kernel sentence about expression. Use a subject and a verb.
Escribe una oración base sobre expresión. Usa un sujeto y un verbo.

2. Sentence Expansion
because · but · so

Kernel: Expression matters in math
Expresión importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Expression matters in math because ____.** *Expresión importa en matemáticas porque ____.*
but *pero*	**Expression matters in math, but ____.** *Expresión importa en matemáticas, pero ____.*
so *entonces*	**Expression matters in math, so ____.** *Expresión importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about expression.
Di un hecho verdadero sobre expression.

Expression ____.

Question
Pregunta

Ask a question about expression.
Haz una pregunta sobre expression.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about expression.
Muestra entusiasmo sobre expression.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with expression.
Dile a un compañero qué hacer con expression.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

I substitute s = 5 into 150 + 12s to get ____.
Sustituyo s = 5 en 150 + 12s para obtener ____.

One expression works for any show because s ____.
Una expresión sirve para cualquier concierto porque s ____.

First I substitute, then I do ____ because of the order of operations.
Primero sustituyo, luego hago ____ por el orden de las operaciones.

Try It

Solve on your own. Check the answer key when you are done.

1. Which expression means '5 less than twice a number n'?

A. 2n – 5

B. 5 – 2n

C. 2(n – 5)

D. 2 – 5n

Show your work:

2. Evaluate 4m + 1 when m = 5.

A. 21

B. 20

C. 9

D. 25

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A gym membership costs $25 per month plus a $40 signup fee. Write an expression using m for months, then evaluate it for 6 months and 12 months. Explain what each part of the expression represents.

Sentence starter: The expression is ____. For 6 months: ____ = ____. For 12 months: ____ = ____. The ____ represents the ____ and the ____ represents the ____.

Show your work:

Reflect — Exit Ticket

Evaluate $5(n - 3) + 2$ when $n = 7$.

A. 22
B. 32
C. 17
D. 36

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $2n - 5$ — *Twice n is $2n$. Five less than that is $2n - 5$.*
2. **Try It 2:** A. 21 — *$4(5) + 1 = 20 + 1 = 21$.*
3. **Exit Ticket:** A. 22 — *$5(7 - 3) + 2 = 5(4) + 2 = 20 + 2 = 22$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Expression is a math phrase with numbers and letters, but no equal sign.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).